SOEN343: SOFTWARE ARCHITECTURE

Smart Urban Mobility Management System (SUMMS) (Phase II)

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1 General Information

- **Weight:** 4% of the overall grade.

2 Introduction

This assignment targets:

1. Understanding the system architecture.
2. Understanding the use case diagram.
3. Understanding the sequence diagram.

3 Your Project (Phase II)

3.1 System Architecture

In this section, you are tasked with selecting a suitable architecture for the system. Provide a justification for your choice by explaining how it aligns with the system's requirements and goals. Highlight the architectural qualities you are prioritizing (such as scalability, maintainability, performance, security, etc.), and discuss how your chosen architecture addresses these concerns. Refer to course materials to help you select an appropriate architectural style (e.g., layered, microservices, event-driven, etc.), and explain the reasoning behind your selection. Identify all system components and their interrelationships. Additionally, compare your selected architecture with at least one alternative, discussing the strengths and trade-offs of each.

3.2 Use Cases

3.2.1 Use Case Diagram

A use case diagram is a graphical depiction of a user who interacts with a system. A use case diagram shows various use cases and different types of users the system has and will often be accompanied by other types of diagrams as well. The system under study includes rental features, analytical features, parking features, and public transportation features. In your use case diagram, you should explicitly represent all functionalities related to the rental system and the analytics system. The other systems (parking and public transportation) should be abstracted as single features, representing the services they provide to the system.

Below are some features that your system may include. You are encouraged to add any additional features that make your system unique:

1. **Subscription and User Access** – applicable to all types of users
2. **Vehicle Search** – including search parameters such as vehicle type, city, and price
3. **Rental Lifecycle** – covering the process from reservation to payment and vehicle return
4. **Vehicle Management (Mobility Provider)** – including vehicle and data management

5. **Access to Other Services** – such as parking and public transportation
6. **Analytics**
    - **Rental Service Analytics:** number of rentals, rentals per city, rentals per mobility type, etc.
    - **Gateway-Level Analytics:** number of accesses to the various services

3.2.2 Use Case Scenarios

For each use case described in the above section, provide a use case scenario describing the behavior of the system. Use the below template for your scenarios:

ID:	**[Unique ID of this use case]**
Title:	[Enter the goal of the use case - preferably as a short, active verb phrase]
Description:	[Describe the goal and context of this use case. This is usually an expanded version of what you entered in the "Title" field.]
Primary Actor:	[A person or a software/hardware system that interacts with your system to achieve the goal of this use case.]
Preconditions:	[Describe the state the system is in before the first event in this use case.]
Postconditions:	[Describe the state the system is in after all the events in this use case have taken place.]
Inputs:	[known inputs for the use case]
Outputs:	[Known outputs of the use case]
Main Success Scenario:	[Describe the flow of events from preconditions to postconditions, when nothing goes wrong.]

Figure 1: Use Case Scenario

3.3 Sequence Diagram

A sequence diagram is an interaction diagram that models the execution of a specific scenario within a system. At this stage, you are required to produce a sequence diagram for the rental lifecycle, representing the complete rental cycle from vehicle search and reservation to payment and vehicle return.

The sequence diagram should provide sufficient detail, based on the requirements phase, to clearly illustrate the control flow and interactions among the system components involved in the rental process.

Before finalizing the design class diagram, it is essential to develop this sequence diagram as part of the design work. The diagram should depict the exchange of method calls between objects, using standard sequence diagram constructs such as opt, alt, and ref, where applicable.

This sequence diagram will support a better understanding of system behavior during the rental process and may be refined or extended in later stages of development.

4 Submission Details

4.1 What to Submit

The whole assignment must be submitted by the due date under the corresponding assignment link. It has to be completed by **all members** of the team, and only **one file** in PDF or WORD format should be submitted.

4.2 Submission Notes

- Clearly include the names and student IDs of all team members in the submission.
- Only one copy of the assignment is to be submitted.
- Ensure that the project is uploaded to the **correct link** on Moodle.
- **No email submissions are accepted.** Projects submitted via email or uploaded to the wrong system or folder will be discarded without resubmission.
- Verify Moodle access before the submission deadline, as **no deadline extensions** will be granted.

4.3 File Naming Convention

The uploaded file should be named using the following convention:

a#_studids.pdf

where:

- # represents the project phase number.
- studids represents the student ID of the team leader.

Example: For Phase 2, if the team leader's student ID is 12345678, the submission file should be named:

a2_12345678.pdf

5 Submission Template

The submitted document should be organized as follows:

5.1 First Page

- Project Name
- Team Name
- Team Members with Student IDs
- Document Purpose (e.g., Phase 2: System Architecture and Sequence Diagram)
- Date

5.2 Second Page

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5.3 Body of the Report

1. **System Architecture**
   Details of the chosen architecture, discussion of its appropriateness for the system, and comparison with an alternative approach.

2. **Use Cases**
   Introduction to the system functionalities, followed by a use case diagram and detailed scenarios.

3. **Sequence Diagram**
   Each functionality should be linked to a real-life scenario, followed by the corresponding sequence diagram. Include detailed description for better understanding.

6 Grading Scheme

Criteria	**Marks**
Quality of the report	0 – 10
System Architecture (discussion and identification of your choice,architecture diagrams)	0 – 20
Use Case Diagram (notation, correct diagrams, complete scenarios)	0 – 20
Sequence Diagram (representation, quality and correctness ,tool usage)	0 – 20
Documentation and Professionalism	0 – 10
Overall Assessment (originality, completeness)	0 – 20
Total	**0 – 100**